

Weighted Hénon prime lifts: analytic domains, Fredholm regions, and a cubic trace obstruction

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Abstract

For the integral area-preserving Hénon map $H_6(x, y) = (1 - 6x^2 - y, x)$, a locked source constructs a local mixing hyperbolic survivor conjugate to a four-state subshift. We extract two all-order consequences. First, if L_o is the unstable multiplier modulus of a primitive survivor orbit, the exact cone expansion $L_o \geq (773/224)^{n_o}$, together with the four-state orbit count, gives a certified zero-free disk and an explicit primitive-period tail for every nonnegative multiplier weight. The Euler/flat correction has a larger analytic disk, but does not continue the complete determinants or certify the reported finite-section root.

Second, we copy the same four-state matrix over all rational primes with one common Dirichlet clock. The resulting direct sum is in S_q exactly when $q \operatorname{Re} s > 1$, has an ordinary Fredholm determinant for $\operatorname{Re} s > 1$, and genuine regularized determinants on their exact Schatten half-planes. Its inverse determinant factors into Riemann zeta functions, but its prime-power trace weights begin $(F_1, F_2, F_3) = (1, 1, 4)$: primes and squares match von Mangoldt, while cubes have exact surplus $3 \log p$. Positive multiplier weights produce a fractional non-meromorphic singularity at $s = 1$; the unique common scalar normalization repairs orders one and two but still fails at cubes.

The local survivor is intrinsic, but copying it over prime labels is an engineered functor rather than a finite-field reduction or canonical global operator. Gates A–E remain false/open. No Hilbert–Pólya operator, Riemann-zero identification, completed-zeta divisor equality, or proof of the Riemann Hypothesis is claimed.

1 Frozen foundation and exact scope

The RH-1–RH-361 corpus remains frozen in four provenance-preserving volumes [4]. RH-362 and RH-363 opened an independent arithmetic-dynamical branch, but left the physical route coordinate

$$\text{actual_same_clock_unnormalized_head_transport_open} \tag{1.1}$$

unchanged [3]. Nothing here estimates the physical same-clock defect, supplies a typed q/E_{off} theorem, or closes the RH-241 moving noisy trace envelope.

The external source studied in this paper is the map

$$H_6(x, y) = (1 - 6x^2 - y, x) \tag{1.2}$$

and the explicitly defined compact survivor Λ_* in four rational state rectangles [5]. On Λ_* , the map is conjugate to the mixing subshift with adjacency matrix

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{1.3}$$

The same source proves an exact period-preserving bijection between primitive symbolic necklaces and primitive Hénon orbits in Λ_* , as well as a two-sided cone certificate. These are theorems about this local survivor, not the full nonwandering set of the Hénon plane.

Write

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \kappa = \frac{773}{224}. \quad (1.4)$$

The numerical finite-volume matrices and the period-eight through period-twelve root table in the source remain finite experiments. We use neither as asymptotic evidence.

2 Symbolic counts and a certified weighted Euler disk

Let $F_n = \text{Tr}(A^n)$ and let p_n be the number of primitive survivor orbits of period n .

Theorem 2.1 (Exact all-order symbolic ledger). *For every $n \geq 1$,*

$$\det(\lambda I - A) = (\lambda^2 - \lambda - 1)(\lambda^2 + 1), \quad (2.1)$$

$$F_n = L_n + 2 \cos\left(\frac{n\pi}{2}\right), \quad (2.2)$$

$$p_n = \frac{1}{n} \sum_{d|n} \mu(d) F_{n/d}, \quad 0 \leq p_n \leq \frac{F_n}{n} \leq \frac{4\varphi^n}{n}, \quad (2.3)$$

where $L_n = \varphi^n + (-\varphi^{-1})^n$ is the n -th Lucas number.

Proof. Direct expansion of $\det(\lambda I - A)$ gives (2.1). The eigenvalues are $\varphi, -\varphi^{-1}, i, -i$, so taking power traces proves (2.2). The period-preserving conjugacy and finite-state fixed-point identity give $F_n = \sum_{d|n} dp_d$; Möbius inversion gives the first part of (2.3). Since the points on primitive period- n orbits form a subset of $\text{Fix}(A^n)$, one has $np_n \leq F_n$. Finally, $|L_n| \leq \varphi^n + 1$ and $|2 \cos(n\pi/2)| \leq 2$, which is at most $4\varphi^n$. $\square$

For a primitive survivor orbit o , let n_o be its primitive period, $\sigma_o \in \{-1, 1\}$ the sign of its unstable multiplier, and $L_o > 1$ its modulus. The cone certificate acts on the invariant unstable line at every iterate and gives

$$L_o \geq \kappa^{n_o}. \quad (2.4)$$

Here L_o is a monodromy multiplier of H_6 ; it is not an eigenvalue of the four-state matrix A .

For real $\beta \geq 0$, define

$$Z_\beta(z) = \prod_{o \in \mathcal{P}_*} (1 - z^{n_o} L_o^{-\beta})^{-1}, \quad D_{E,\beta}(z) = Z_\beta(z)^{-1}. \quad (2.5)$$

Theorem 2.2 (Certified weighted disk and all-order tail). *For every real $\beta \geq 0$, both objects in (2.5) converge normally, are holomorphic, and are zero-free on*

$$|z| < R_\beta := \frac{\kappa^\beta}{\varphi}. \quad (2.6)$$

Fix $0 < R < R_\beta$, put $t = R\kappa^{-\beta}$ and $q = \varphi t$. If the logarithm is truncated after all primitive periods at most N , then

$$\sup_{|z| \leq R} \left| \log Z_\beta(z) - \log Z_\beta^{(\text{prim} \leq N)}(z) \right| \leq \frac{4q^{N+1}}{(N+1)(1-q)(1-t^{N+1})}. \quad (2.7)$$

Proof. For an orbit of period n , equation (2.4) bounds its logarithmic repetition sum by $t^n/(1-t^n)$ on the closed disk. Theorem 2.1 therefore gives

$$\sum_{n>N} p_n \frac{t^n}{1-t^n} \leq \frac{4}{(N+1)(1-t^{N+1})} \sum_{n>N} (\varphi t)^n,$$

which is (2.7). Normal convergence supplies a canonical holomorphic logarithm. Exponentiating it and its negative proves the zero-free statements. $\square$

This is a certified disk, not a maximal-domain theorem. In particular, for $\beta = 1$,

$$R_1 = \frac{\kappa}{\varphi} = 2.132769077248\dots, \quad (2.8)$$

while the source reports a finite-section reciprocal root near 3.429. That numerical value lies outside the proved disk.

2.1 The larger flat correction

The periodic-point flat determinant has the formal factorization

$$D_F(z) = \prod_{o \in \mathcal{P}_*} \prod_{j \geq 0} \left(1 - \sigma_o^j L_o^{-(j+1)} z^{n_o}\right)^{j+1}. \quad (2.9)$$

Its $j = 0$ factor is $D_{E,1}$. Define the remaining correction

$$C_F(z) = \prod_{o \in \mathcal{P}_*} \prod_{j \geq 1} \left(1 - \sigma_o^j L_o^{-(j+1)} z^{n_o}\right)^{j+1}. \quad (2.10)$$

Theorem 2.3 (Analytic flat correction, with a strict firewall). *The product C_F converges normally and is holomorphic and zero-free on*

$$|z| < R_F^{\text{corr}} := \frac{\kappa^2}{\varphi} = 7.359957574612\dots \quad (2.11)$$

The analytic identity $D_F = D_{E,1} C_F$ is directly licensed by the products only on the common smaller disk $|z| < \kappa/\varphi$. The larger correction disk alone does not continue either complete determinant.

Proof. For $0 < x < 1$,

$$\sum_{j \geq 1} (j+1)x^{j+1} = \frac{x^2(2-x)}{(1-x)^2}. \quad (2.12)$$

Use $x = L_o^{-r}$, sum the logarithmic repetitions, and apply (2.4). Uniformly in n_o and r , the right side of (2.12) is bounded by a constant times κ^{-2rn_o} . The remaining orbit sum is dominated by $\sum_n 4\varphi^n u^n/(1-u^n)$ with $u = |z|\kappa^{-2}$, which converges when $\varphi u < 1$. Each factor is then nonzero, so the normal product is zero-free. The final scope statement follows because theorem 2.2 controls the $j = 0$ Euler factor only to κ/φ . $\square$

3 A common-clock prime-copy operator

Let ℓ denote a rational prime and use the real logarithm in $\ell^{-s} = e^{-s \log \ell}$. On $\mathcal{H} = \bigoplus_{\ell} \mathbb{C}^4$, define

$$\mathcal{T}_s = \bigoplus_{\ell \text{ prime}} \ell^{-s} A. \quad (3.1)$$

This is a common-clock copy: every prime receives the same local matrix and the same exponent s . It is not a reduction of H_6 modulo ℓ . We use the standard Schatten and regularized-determinant conventions of Simon [1].

Theorem 3.1 (Exact bounded, compact, and Schatten regions). *The operator $\mathcal{T}_s$ is bounded exactly for $\operatorname{Re} s \geq 0$, and compact exactly for $\operatorname{Re} s > 0$. For every finite $q > 0$,*

$$\|\mathcal{T}_s\|_{S_q}^q = \|A\|_{S_q}^q \sum_{\ell} \ell^{-q \operatorname{Re} s}, \quad \mathcal{T}_s \in S_q \iff q \operatorname{Re} s > 1. \quad (3.2)$$

Proof. The norm of the ℓ -block is $\ell^{-\operatorname{Re} s} \|A\|$. These block norms are uniformly bounded exactly for $\operatorname{Re} s \geq 0$, and tend to zero exactly for $\operatorname{Re} s > 0$. Each singular value of A is multiplied by $\ell^{-\operatorname{Re} s}$, so summing their q -powers gives (3.2). The prime Dirichlet series converges exactly when its real exponent exceeds one. $\square$

The unweighted survivor determinant is

$$D_{E,0}(z) = \det(I - zA) = 1 - z - z^3 - z^4 = (1 + z^2)(1 - z - z^2). \quad (3.3)$$

Theorem 3.2 (Fredholm identity and scalar zeta factorization). *For $\operatorname{Re} s > 1$,*

$$\mathcal{D}_A(s) := \det_F(I - \mathcal{T}_s) = \prod_{\ell} (1 - \ell^{-s} - \ell^{-3s} - \ell^{-4s}) \quad (3.4)$$

$$= \prod_{n \geq 1} \zeta(ns)^{-p_n}. \quad (3.5)$$

Consequently

$$Z_A(s) := \mathcal{D}_A(s)^{-1} = \prod_{n \geq 1} \zeta(ns)^{p_n} \quad (3.6)$$

in the same region. The scalar right sides continue meromorphically to

$$\operatorname{Re} s > \log_2 \varphi, \quad (3.7)$$

but below $\operatorname{Re} s > 1$ they are not ordinary Fredholm determinants of $\mathcal{T}_s$.

Proof. Theorem 3.1 makes $\mathcal{T}_s$ trace class for $\operatorname{Re} s > 1$. The Fredholm determinant then factors normally over the finite-dimensional blocks, and (3.3) gives (3.4). The primitive-orbit identity $\det(I - zA) = \prod_n (1 - z^n)^{p_n}$ and absolute convergence permit interchange of the prime and period products, giving (3.5). The sign is essential: positive zeta exponents belong to the inverse determinant.

For a compact set in $\operatorname{Re} s > \log_2 \varphi$, the large- n estimate $|\log \zeta(ns)| \ll 2^{-n \operatorname{Re} s}$ combines with $p_n \leq 4\varphi^n/n$. The tail product therefore converges normally; finitely many initial zeta factors are meromorphic. $\square$

Theorem 3.3 (Exact regularized determinant regions). *Let $m \geq 1$ be an integer. On $m \operatorname{Re} s > 1$, the regularized determinant exists and has the branch-free block product*

$$\det_m(I - \mathcal{T}_s) = \prod_{\ell} \left[(1 - \ell^{-s} - \ell^{-3s} - \ell^{-4s}) \exp \left(\sum_{r=1}^{m-1} \frac{F_r}{r} \ell^{-rs} \right) \right]. \quad (3.8)$$

In particular $\det_2(I - \mathcal{T}_s)$ is a genuine regularized determinant on $\operatorname{Re} s > 1/2$.

Proof. Theorem 3.1 gives $\mathcal{T}_s \in S_m$ exactly in the stated region. In finite dimension,

$$\det_m(I - zA) = \det(I - zA) \exp \left(\sum_{r=1}^{m-1} \frac{z^r \operatorname{Tr}(A^r)}{r} \right). \quad (3.9)$$

The defining regularized product converges normally because its local remainder begins at order m , and $\sum_{\ell} \ell^{-m \operatorname{Re} s} < \infty$. Equations (3.3) and (3.9) give (3.8). $\square$

Remark 3.4. The power-trace logarithm $-\sum_{r \geq m} F_r \sum_{\ell} \ell^{-rs}/r$ is directly absolutely convergent only after also imposing $\operatorname{Re} s > \log_2 \varphi$. The block product (3.8), not an unlicensed logarithm branch, is the global statement on the full S_m half-plane.

Corollary 3.5 (An entropy-block zero of the regularized determinant). *At the positive real point*

$$s_2 = \log_2 \varphi = 0.694241913630 \dots, \quad (3.10)$$

the operator belongs to S_2 and $\det_2(I - \mathcal{T}_{s_2}) = 0$.

Proof. The $\ell = 2$ block has scalar $2^{-s_2} = \varphi^{-1}$, and (3.3) vanishes at $z = \varphi^{-1}$. The regularizing exponential is nonzero. Also $2s_2 > 1$, so theorem 3.3 applies. $\square$

This zero is the topological-entropy zero of one copied $\ell = 2$ block. It is not a Riemann zero, a self-adjoint eigenvalue, or evidence for Gate E.

4 The first arithmetic defect is at prime cubes

In $\operatorname{Re} s > 1$, differentiate the inverse determinant logarithm:

$$-\frac{\mathcal{Z}'_A}{\mathcal{Z}_A}(s) = \sum_{\ell} \sum_{r \geq 1} F_r \log \ell \ell^{-rs}. \quad (4.1)$$

The classical von Mangoldt ledger would have coefficient one at every power ℓ^r .

Theorem 4.1 (Exact cubic trace obstruction). *The first symbolic ledgers are*

$$(F_1, F_2, F_3) = (1, 1, 4), \quad (p_1, p_2, p_3) = (1, 0, 1). \quad (4.2)$$

Thus (4.1) gives the classical weight $\log \ell$ at primes and prime squares, but gives $4 \log \ell$ at prime cubes. The first surplus is exactly $3 \log \ell$.

Proof. Theorem 2.1 gives $F_1 = 1, F_2 = 1, F_3 = 4$. Möbius inversion gives $p_1 = 1, p_2 = (F_2 - F_1)/2 = 0$, and $p_3 = (F_3 - F_1)/3 = 1$. Substitution in (4.1) proves the weight statement. $\square$

Equivalently, the first zeta factors of $\mathcal{Z}_A$ are

$$\mathcal{Z}_A(s) = \zeta(s) \zeta(3s) \zeta(4s)^2 \dots. \quad (4.3)$$

The factor $\zeta(s)$ comes from the unique survivor fixed orbit. The first chaotic surplus is the unique primitive period-three orbit and the factor $\zeta(3s)$. The appearance of Riemann zeta here is forced by copying the same finite-state orbit factor over the prime labels; it is not spectral recovery of Riemann zeros.

5 Positive weights and common scalar normalization

The full real Hénon map has two fixed points. Only one belongs to the certified survivor.

Lemma 5.1 (The unique survivor fixed atom). *The unique fixed point in Λ_* is*

$$P_* = (x_*, x_*), \quad x_* = -\frac{1 + \sqrt{7}}{6}. \quad (5.1)$$

Its unstable multiplier modulus is

$$L_* = 1 + \sqrt{7} + \sqrt{7 + 2\sqrt{7}} = 7.151675243800 \dots. \quad (5.2)$$

Proof. The fixed-point equation is $6x^2 + 2x - 1 = 0$. The negative root lies in the negative state interval; the positive root does not lie in either certified state interval. At the negative root, the derivative has determinant one and trace $-12x_* = 2 + 2\sqrt{7}$. Solving its quadratic characteristic equation gives (5.2). $\square$

For real $\beta \geq 0$ and $c > 0$, define, initially in a sufficiently far right half-plane,

$$\mathfrak{Z}_{\beta,c}(s) = \prod_{\ell} Z_{\beta}(c\ell^{-s}). \quad (5.3)$$

Theorem 5.2 (Natural weighted lift and a fractional singularity). *For $c = 1$ and every real $\beta \geq 0$, the product (5.3) converges normally and is holomorphic and zero-free on $\operatorname{Re} s > 1$. As real $\sigma \downarrow 1$,*

$$\log \mathfrak{Z}_{\beta,1}(\sigma) = L_*^{-\beta} \log \frac{1}{\sigma - 1} + O(1). \quad (5.4)$$

For $\beta = 0$, the singularity is an ordinary simple pole. For every real $\beta > 0$, the exponent lies strictly between zero and one, so $\mathfrak{Z}_{\beta,1}$ has no meromorphic continuation through $s = 1$. Along the real boundary, its inverse has corresponding fractional-order vanishing; this is not a holomorphic zero through $s = 1$.

Proof. Theorem 2.2 gives $R_{\beta} \geq 1/\varphi > 1/2$. Thus all local arguments ℓ^{-s} stay in one compact subdisk for $\operatorname{Re} s \geq 1$, and

$$\log Z_{\beta}(z) = L_*^{-\beta} z + O(z^2) \quad (5.5)$$

uniformly there. Summing the linear term over primes gives $L_*^{-\beta} \sum_{\ell} \ell^{-s}$, while the quadratic remainders converge normally near $s = 1$. The prime zeta asymptotic [2] $\sum_{\ell} \ell^{-\sigma} = \log(1/(\sigma - 1)) + O(1)$ proves (5.4). A meromorphic function has an integer zero or pole order, so a noninteger leading logarithmic coefficient is impossible. $\square$

To test whether one common scalar can repair the arithmetic trace weights, put

$$G_m(\beta) = \sum_{o:n_o|m} n_o L_o^{-\beta m/n_o}, \quad Q_m(\beta, c) = c^m G_m(\beta). \quad (5.6)$$

Whenever (5.3) is expanded in a common convergence half-plane,

$$-\frac{\mathfrak{Z}'_{\beta,c}(s)}{\mathfrak{Z}_{\beta,c}(s)} = \sum_{\ell, m \geq 1} Q_m(\beta, c) \log \ell \ell^{-ms}. \quad (5.7)$$

Theorem 5.3 (Scalar normalization fails exactly at cubes). *Fix real $\beta \geq 0$. If $c > 0$ is required to match the prime weight, $Q_1(\beta, c) = 1$, then uniquely*

$$c = L_*^{\beta}. \quad (5.8)$$

For this scalar,

$$Q_2(\beta, L_*^{\beta}) = 1, \quad Q_3(\beta, L_*^{\beta}) = 1 + 3 \left(\frac{L_*^3}{L_3} \right)^{\beta} > 1, \quad (5.9)$$

where L_3 is the unstable multiplier modulus of the unique primitive period-three survivor orbit. At $\beta = 0$, the cubic value is four. Therefore no common positive scalar normalization produces von Mangoldt weights at all prime powers.

Proof. Only the fixed orbit divides order one, so $Q_1 = cL_*^{-\beta}$, proving (5.8). Since $p_2 = 0$, order two contains only the second repetition of the fixed orbit and hence $Q_2 = c^2 L_*^{-2\beta} = 1$. Since $p_3 = 1$, order three contains the third fixed repetition and one primitive period-three orbit, giving $G_3 = L_*^{-3\beta} + 3L_*^{-\beta}$. Multiplication by $c^3 = L_*^{3\beta}$ gives (5.9). $\square$

The coefficient obstruction is valid for every real $\beta \geq 0$. The analytic normalized product near $s = 1$ requires an additional strict radius condition.

Proposition 5.4 (Exact currently certified normalization threshold). *The all-order source bound proves that the normalized local arguments in (5.3), with $c = L_*^\beta$, lie strictly inside the weighted Euler disk at $s = 1$ exactly when*

$$0 \leq \beta < \beta_0, \quad \beta_0 = \frac{\log(2/\varphi)}{\log(L_*/\kappa)} = 0.290834898770\dots \quad (5.10)$$

In this range the normalized prime lift has a simple zeta pole times a holomorphic nonzero factor near $s = 1$. For $\beta \geq \beta_0$, the exact coefficient identities (5.8)–(5.9) remain valid, but the present source does not license the same near- $s = 1$ infinite-product statement.

Proof. At $s = 1$, the largest local argument is the $\ell = 2$ argument $L_*^\beta/2$. Theorem 2.2 places it strictly inside the certified disk precisely when $L_*^\beta/2 < \kappa^\beta/\varphi$, which is (5.10). In the strict range, the local logarithm has first coefficient one and a uniformly summable quadratic remainder, so the global product is $\zeta(s)$ times a holomorphic nonzero factor. Equality or reversal of the radius inequality supplies no such continuation theorem. $\square$

Complex and negative weights are outside the theorem. Because every $L_o > 0$, complex powers can be defined using the real logarithm, but the absolute-value estimates then depend on $\operatorname{Re} \beta$, and Q_3 has no real order relation. Negative real weights require an explicit all-order multiplier upper envelope that is not established in this paper.

6 Route A value, Route B obstruction, and Gates

Route	Exact verdict
Route A	GO. The local survivor yields an all-order weighted analytic theorem and a larger flat correction domain. The prime-copy functor has exact compact/Schatten/Fredholm regions, a scalar zeta factorization, a fractional singularity theorem, and a strict cubic normalization obstruction.
Route B	STOP_SCOPED. The first fatal mismatch is Gate A data type: copying one local matrix over all prime labels is engineered, not a canonical arithmetic globalization. Even if this functor is accepted, Gate D fails at prime cubes because the trace weight is four rather than one.

The following distinctions are mandatory.

1. The survivor Λ_* is an intrinsic local hyperbolic set, but $\mathcal{T}_s$ is not the reduction of that real set modulo primes. The prime labels are added by definition.
2. Neither $\mathcal{D}_A$ nor $\det_2(I - \mathcal{T}_s)$ is the stationary-centered physical noisy determinant used in the original RH route.
3. The factorization into $\zeta(ns)$ is an exact consequence of copying the same primitive orbit factors over prime labels. It is not a discovery of Riemann zeros in the Hénon spectrum.
4. The cube obstruction excludes common scalar normalization of this positive orbit ledger. It does not exclude signed cancellations, orbit-dependent weights, or a different intrinsic global construction.

5. The larger flat correction disk does not certify the source’s finite root near 3.429, and finite root stability is not analytic continuation.

Thus the Gate ledger remains

Gate	Status	First missing or failed object
A	false/open	canonical intrinsic global physical determinant
B	false/open	time-oriented scattering or unitary completion
C	false/open	self-adjoint generator and intrinsic $T \log T$
D	false/open	cube weight is $4 \log p$, not $\log p$
E	false/open	completed-zeta divisor equality

No Hilbert–Pólya operator, self-adjoint generator, Riemann-zero spectral identification, completed-zeta divisor equality, or proof of the Riemann Hypothesis follows.

7 Executable protocol

The artifact performs finite exact reproduction and provenance checks only:

1. it hashes the locked survivor theorem, contraction, weighted-zeta, orbit, limitation, RH-363, and four-volume sources;
2. it recomputes $F_1, \dots, F_{12}$, the Lucas trace identity, and $p_1, \dots, p_{12}$;
3. it checks $\det(I - zA) = 1 - z - z^3 - z^4$, the exact fixed point and multiplier, the two certified radii, and β_0 ;
4. it verifies the source contains exactly one selected survivor orbit at periods one and three and reproduces representative scalar-normalization ledgers;
5. it requires every Gate and every forbidden macro claim to remain false.

These rows do not prove the all-order convergence theorems; sections 2–5 do. They also do not promote the stored multiplier decimals to interval certificates beyond the exact symbolic multiplicities and analytic formulas used in the proofs.

8 Conclusion

The certified survivor supplies more than a finite cycle table: its exact symbolic entropy and cone expansion give a rigorous weighted analytic domain. Prime copying then yields unusually clean operator ideals and Fredholm formulas. Precisely because the construction is clean, its arithmetic failure is visible at the first nontrivial primitive orbit: the unique period-three orbit changes the cube trace weight from one to four.

This is useful Route A structure and a sharp Route B boundary. The natural next questions are a positive-radius theorem for the unweighted return-cycle bouquet, a genuine primitive-divisor theorem for its coefficient anchors, or an intrinsic globalization that does not add prime copies by hand. The original physical same-clock route remains open and unchanged.

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